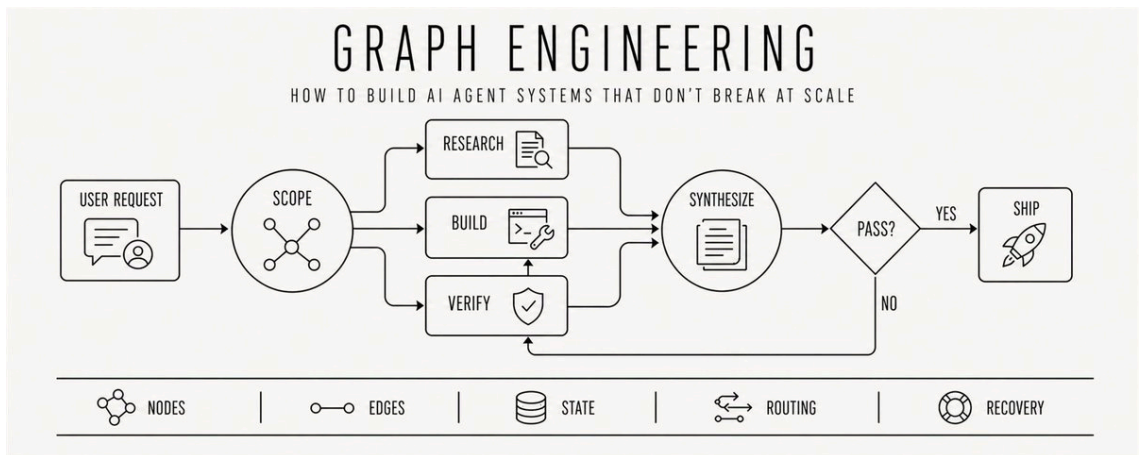




← Post



rari 
@0xwhrrari



Graph Engineering: How to Build AI Agent Systems That Don't Break at Scale

Most builders still design AI agents as a straight line

Research first

Write second

Review third

Ship last

Each step waits for the one before it, even when half of them never needed the previous result

The system does not branch

It does not parallelize

indow until the agent gets slow, confused, or

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What graph engineering actually means

Graph engineering is the practice of turning an agentic workflow into an explicit execution map

Instead of hiding every decision inside one model loop, you define the system as nodes and edges

text



```

NODE    = one bounded unit of work
EDGE    = a real dependency between two nodes
STATE   = the data that survives between nodes
ROUTER  = the rule that selects the next edge
GATE    = the check that decides whether work can continue
```

A node can be an agent, a tool call, a deterministic function, a verifier, or a human approval step

An edge says what is allowed to run next and what data crosses the boundary

The graph decides which loops run, in what order, with which branches, joins, and recovery paths

A loop helps one agent improve its work A graph coordinates many loops into one system

1. Stop treating every "and then" as a dependency

Most agent workflows are linear because that is how people write instructions

Do A, then B, then C

But sequence is not the same as dependency

If B does not consume A's output, there is no reason for B to wait



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ject repository -> check competitor price

```
USER REQUEST    -> collect market data -----
                -> inspect repository -----> SYNTHESIZE
                -> check competitor pricing ----
```

The first question in graph engineering is simple

Does the next step actually read the previous step's output

If the answer is no, cut the edge

That single change usually turns a slow chain into a fast parallel graph

2. Give every node a contract

A node you cannot describe precisely is a node you cannot route, test, or replace

Every useful node needs four things

- One job
- Explicit input
- Structured output
- A clear failure state

text



```
{
  "node": "source_researcher",
  "input": {
    "topic": "string",
    "source_type": "primary"
  },
  "output": {
    "claim": "string",
    "source_url": "string",
    "confidence": "high | medium | low"
  },
  "failure": "no primary source_found"
```

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less what happened

sponse into something the graph can trust

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You can swap the model, prompt, or tool without rebuilding the entire system as long as the contract stays the same

3. Treat edges as data contracts, not arrows

An edge should not mean "B comes after A"

It should mean "A produced data that B is allowed to consume"

That distinction matters because most workflow plumbing does not need another model call

Flattening arrays, removing duplicates, filtering nulls, checking a status, and joining records are deterministic operations

text



```
const usable = results
  .filter(Boolean)
  .flatMap(result => result.items)

const unique = [...new Map(
  usable.map(item => [item.source_url, item])
).values()]
```

No agent is needed here

Save model calls for judgment

Use code for plumbing

A graph where every edge is another agent is paying tokens for its own wiring

4. Learn the four shapes behind almost every agent graph

You do not need fifty patterns

variations of four shapes

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requires the previous output

It is simple, predictable, and often slower than necessary

The diamond

text



```
-> B1 -  
A ->   -> B2 --> C  
      -> B3 -
```

Split one job into independent branches, run them together, then merge the results

This is the workhorse for research, code review, due diligence, and market scans

The router

text



```
      -> QUICK PATH  
CLASSIFY ---  
      -> FULL AUDIT
```

Inspect state and choose only the path the task needs

Small work stays cheap

Risky work gets a deeper graph

The controlled cycle

text



```
WORK -> VERIFY -> PASS -> EXIT  
      |  
      -> FAIL -> FEEDBACK -> WORK
```

e result is incomplete

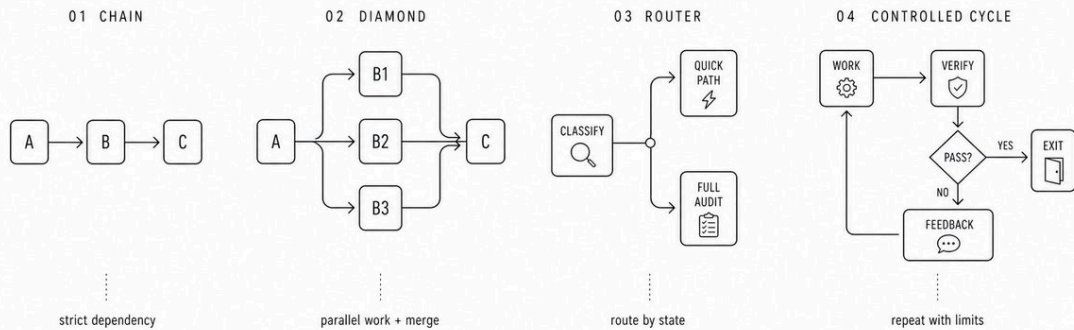
dget, and a convergence rule

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Four Shapes Behind Every Agent Graph

Most agent systems reduce to four reusable structures



5. Fan out independent work, then join it deliberately

Parallelism is the easiest graph advantage to understand and the easiest one to abuse

If five nodes are independent, run them together

text



```
const settled = await Promise.allSettled(
  sources.map(source => researchNode(source))
)

const findings = settled
  .filter(result => result.status === "fulfilled")
  .map(result => result.value)
```

One failed branch should not destroy the other four

But do not place a barrier after every stage

A join is worth the wait only when the next node needs the complete set

Examples include cross-source deduplication, ranking all candidates, comparing coverage is complete

lently, keep the graph streaming

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astYour topology decides where the system

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ble

The graph should enforce what that judgment is allowed to trigger

text



```
const decision = await classifyRisk(change)

switch (decision.severity) {
  case "low":
    return quickReview(change)

  case "high":
    return fullParallelAudit(change)

  default:
    return humanReview(change)
}
```

The classifier is probabilistic

The allowed routes are deterministic

This gives you the model's flexibility without giving it unlimited control over the system

OpenAI's visual Agent Builder makes this shift obvious: agent behavior is increasingly designed as an inspectable workflow instead of a hidden chain of prompts

<https://x.com/i/web/status/1975269388195631492>

7. Put verification on the edge

The highest-leverage node in a graph is often the one that produces nothing new

oving downstream

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ce
orts the claim

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requested schema
path reaches the same conclusion

GENERATOR -> VERIFIER -> PASS -> SYNTHESIZER
|
-> FAIL -> REPAIR

Do not ask the same agent to generate, approve, and publish its own work in one context

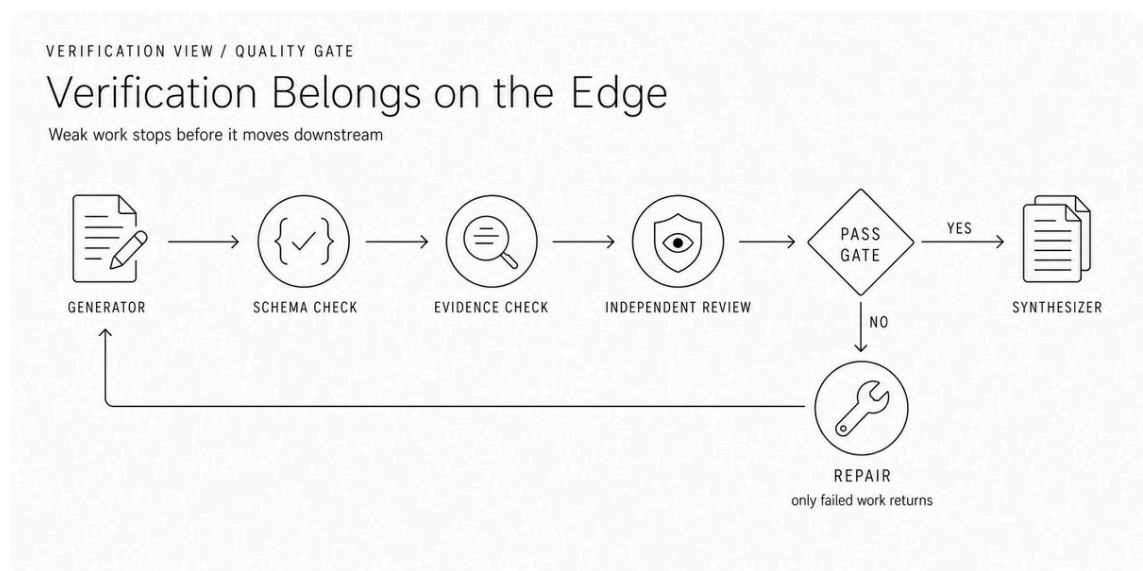
Separate the roles

Separate the prompts

Separate the failure boundaries

Anthropic's production research system follows this logic at a larger scale: a lead agent coordinates parallel subagents, findings are synthesized, and a dedicated citation stage attaches evidence before the result reaches the user

<https://x.com/i/web/status/2041927687460024721>



diagrams hide

he system has to resume after a crash

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state

task_id
current_node
completed_nodes
artifacts
decisions
evidence
budgets
retry_counts
human_approvals

Do not move giant transcripts between nodes

Move references to artifacts

A research node should store its report and return a path, ID, or structured summary

A reviewer should read the artifact directly instead of receiving a compressed retelling through three agents

This reduces context loss and makes every transition auditable

The graph should be able to answer three questions at any moment

text



What has already happened
Why did the system choose this route
Where can execution safely resume

If it cannot answer them, the graph is still a demo

9. Add cycles only when they converge

A cycle is useful when the amount of work is unknown in advance

Recovery, repair, and iterative repair are good examples

condition

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```
const seen = new Set()

while (dryRounds < 2 && iteration < 6) {
  const findings = await discover()
  const fresh = findings.filter(item => !seen.has(item.key))

  fresh.forEach(item => seen.add(item.key))
  dryRounds = fresh.length === 0 ? dryRounds + 1 : 0
  iteration += 1
}
```

Notice what the system remembers

It deduplicates against everything already seen, not only the findings that passed verification

Otherwise rejected ideas keep returning and the graph pays to rediscover the same dead ends forever

Every controlled cycle needs

- A completion test
- A maximum number of rounds
- A token or cost budget
- A record of previous attempts
- An escalation path when convergence fails

10. Design failure as a local event

In a chain, one broken step can freeze the whole workflow

In a graph, failure should stay inside the smallest possible boundary

Each node needs a policy

text



```
DETDRV transient tool or network failure
or source unavailable
failed
validation
entity crossed a threshold
or permission boundary reached
```

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Make writes idempotent so a retry does not duplicate side effects

Give parallel workers isolated workspaces when they modify files

Record every routing decision with the state that produced it

Reliability does not come from hoping every node succeeds

It comes from deciding what the graph does when one does not

11. Topology is your cost model

A graph is not automatically cheaper than one agent

It can burn far more tokens if every task spawns a fleet

The shape controls both latency and cost

Use cheaper models for bounded extraction, classification, and formatting

Use stronger models for decomposition, synthesis, and difficult verification

Route simple tasks through a short path

Reserve the full graph for work that earns it

text



SIMPLE REQUEST -> SMALL MODEL -> QUICK CHECK -> DONE

COMPLEX REQUEST -> PLANNER -> PARALLEL SPECIALISTS

-> VERIFIERS

-> STRONG SYNTHESIZER

-> HUMAN GATE

Anthropic reports that multi-agent research can materially outperform a single agent on breadth-first work, but it also uses far more tokens

That is the tradeoff

minimizing the number of agents

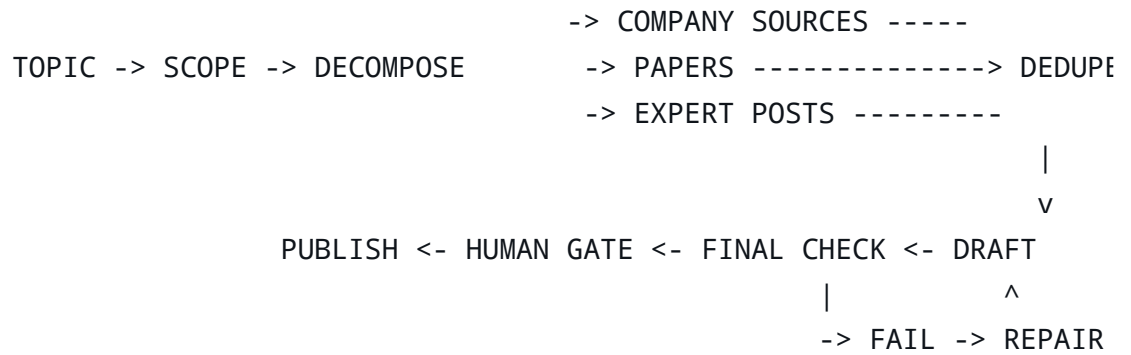
only where parallelism, specialization, or
rough value

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research and publishing

one idea into a cited article

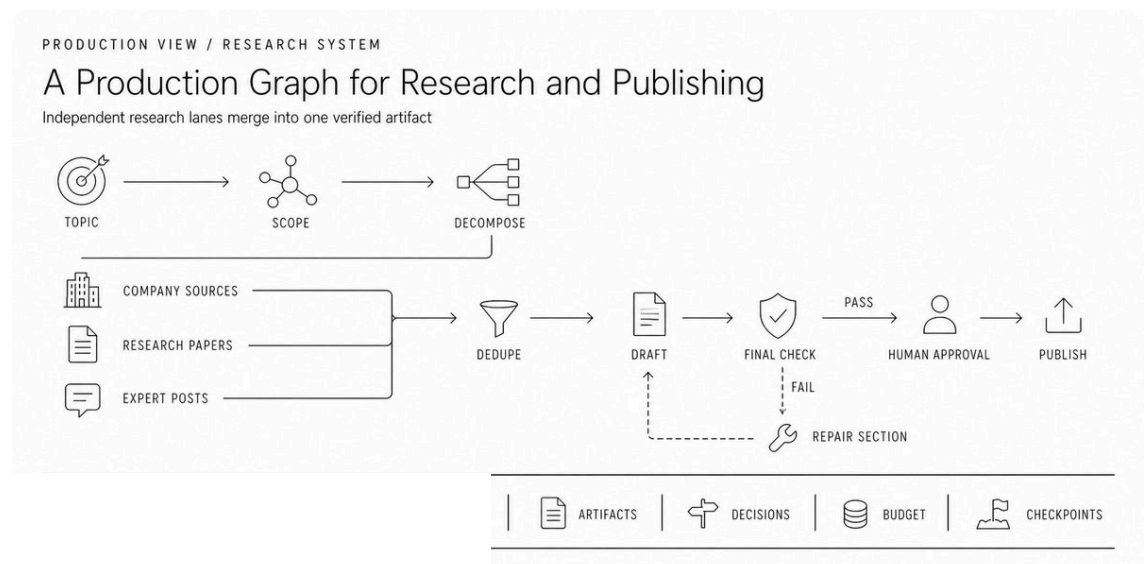


The system works like this

1. The scope node defines the question, audience, and completion criteria
2. The decomposition node creates independent research lanes
3. Research nodes run in parallel with separate contexts
4. Deterministic code removes duplicates and normalizes sources
5. The draft node writes from structured evidence
6. The checker validates claims, citations, style, and missing sections
7. Failed checks route only the relevant section back to repair
8. A human approves the final artifact before publishing

This is not one giant agent pretending to be a team

It is a system with explicit ownership, state, and authority



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chitecture diagram

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- The task is short
- One context can hold all relevant information
- There are no independent branches
- Failure is cheap
- A human can review the final result quickly

Move to a graph when

- Work can run in parallel
- Different nodes need different tools or permissions
- Outputs require independent verification
- The task must resume after interruption
- Several loops need shared state
- Cost and authority must be controlled by route

Start with one loop

Draw a graph only when the dependencies force you to

The graph engineering checklist

Before you ship, ask

text



- [] Does every edge carry real data or authority
- [] Does every node have one bounded job
- [] Are inputs and outputs structured
- [] Can independent nodes run in parallel
- [] Are joins placed only where the full set is required
- [] Are important results verified before moving downstream
- [] Can failures be retried without duplicating side effects
- [] Can the graph resume from a checkpoint
- [] Does every cycle have a hard stop and budget
- [] Can a human interrupt high-risk paths
- [] Can you explain why every route was selected
- [] Is the graph simpler than the problem it solves

no, delete nodes

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instruction

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the model sees

ironment around the model

Loop engineering makes one unit of work improve through feedback

Graph engineering coordinates the entire job

text



PROMPT -> CONTEXT -> HARNESS -> LOOP -> GRAPH
message memory machine run coordination

The model is only one node

The product is the system around it

A prompter asks the agent to do more An architect redesigns the graph so the system can do more safely

Keep reading

- [The Three Layers Behind Reliable AI Agents: Harness vs Loop vs Graph Engineering](#)
- [How I set up Obsidian + Claude as my second brain](#)
- [How I set up Claude to actually get work done](#)
- [How I set up Claude projects so they actually work](#)
- [30 Claude system prompts I actually use](#)
- [Loop Engineering: The AI skill every builder needs in 2026](#)
- [30 Claude Code settings, shortcuts & workflows most users miss](#)
- [How I Use Claude Cowork to Run Like a One-Person Company](#)

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use the patterns later

actical AI systems

ill forcing every agent through one long chain

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OxSlyth   @0xSlyth · 8月10日



legendary

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2

4258



Hussain Hashim | Building SundayBack  @itsthedonh... · 8月10日



@Oxwhrrari yeah, skipping the linear steps is a game changer. I used to spend ages on review when the first draft wasn't even close to done. Total time sink.

1



1

1758



slash1s   @slash1sol · 8月10日



new alpha 🙄🙄🙄

ty bro

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